

3. B.

No.

We should not expect a high probability of exactly finding $\hat{z}_i = z_i$ for every i .

For any fixed P_m and P_n , the number of physical and syndrome error grows as $O(n)$.

The decoder can only correct errors as long as

$$\frac{1}{n} \sum \hat{z}_i z_i > 0$$

This allows $\hat{z}_i \neq z_i$ for some bits.

e.g.

original :	+	+	+	+	+	+	+	...	...	...	...	...	...	...
errored :	+	+	-	-	+	+	+	...	...	...	...	...	...	...
syndrome :	+	+	-	-	+	+	+	...	...	...	...	...	...	...

if we correct these 2 errors then the state from our perspective is corrected but there is still one $\hat{z}_i \neq z_i$.